HFT Tick-to-Trade FPGA

Issues & Resolutions Log

Abhishek

Target: Xilinx Alveo U200 (`xcu200-fsgd2104-2-e`)
Tooling: Vivado 2025.2 Host: Ubuntu 24.04, 18 GB RAM (VM)

Last updated: 12 June 2026

Contents

1. Overview

This document tracks the engineering issues encountered while bringing up the tick-to-trade pipeline (`itch_parser` → `order_book_m2` → `strategy_imbalance` → `latency_counter`) through Xilinx Vivado synthesis. Each entry records the **symptom**, the **evidence**, the **root cause**, and the **resolution / status**.

#	Issue	Severity	Status
1	Synthesis exhausts RAM; order book not inferring BRAM	[HIGH]	**Fixed** (distributed RAM)
2	Vivado native crash during synthesis (`hs_err_pid`)	[HIGH]	Resolved by #1
3	XDC constraint errors in `top.xdc` (lines 31–32)	[MEDIUM]	**Fixed**
4	Assorted synthesis warnings (set/reset, unconnected ports)	[LOW]	Cosmetic / review

2. Issue #1 — Synthesis exhausts RAM; order book stored as flip-flops instead of BRAM [HIGH]

2.1 Symptom

A batch synthesis run (`synth_design -top tick_to_trade_top -part xcu200-fsgd2104-2-e`) ran the VM out of memory. GNOME System Monitor showed **18.0 GB of 18.4 GB (97.7%) used**, swap climbing past 2.3 GB, and the disk pinned at ∼560 MB/s (swap thrashing, not useful work). The run was manually interrupted.

2.2 Evidence

From `Vivado/synth_batch.log`:

```
WARNING: [Synth 8-11357] Potential Runtime issue for 3D-RAM or RAM from
        Record/Structs for RAM entries_reg with 33280 registers
...
2 Input 1 Bit Muxes := 2218
5 Input 1 Bit Muxes := 521
...
Finished Cross Boundary and Area Optimization : elapsed = 00:14:51 free physical = 5393
Finished Applying XDC Timing Constraints : free physical = 1690
Start Timing Optimization
INFO: [Common 17-41] Interrupt caught. Command should exit soon.
wait_on_runs: ... free physical = 167 <- 167 MB RAM left
```

A single optimization phase ran for ∼15 minutes, and free physical RAM fell from ∼10 GB to **167 MB** before the interrupt.

2.3 Root cause

The order book stores 256 orders, each a 130-bit packed struct ($256 \times 130 = 33{,}280$ bits). Vivado mapped this to **33,280 individual flip-flops** plus a large read-multiplexer tree (the 2,218 + 521 muxes above) instead of a Block RAM. Building and optimizing that fabric is what caused the runtime and memory blow-up.

The array fails BRAM inference because the reset branch clears every element at once, which Block RAM physically cannot do:

```
// rtl/order_book_m2.sv : 107-108 (inside the asynchronous reset)
for (int i = 0; i < ORDER_DEPTH; i++)
    entries[i] <= '0;
```

Seeing a requirement to reset all contents simultaneously, the synthesizer falls back from BRAM to flip-flops.

2.4 Resolution / Status (**fixed** 12 June 2026)

Applied to `rtl/order_book_m2.sv`:

1. Removed the `valid` field from the `order_entry_t` struct and the whole-array async reset loop, so the payload array `entries[]` no longer needs clearing on reset.
2. Added a separate resettable bit-vector `logic [ORDER_DEPTH-1:0] entry_valid;` (256 FFs) — a slot's contents are trusted only when its valid bit is set, so the payload never needs a reset.
3. Kept reads *combinational* (0-cycle) so no read latency is added to this latency-critical pipeline. The payload therefore infers **distributed RAM (LUTRAM)** rather than block RAM; true BRAM was declined because it would require synchronous reads and an FSM pipeline change (+1 clock per access). See trade-off note below.

Effect: storage drops from 33,280 flip-flops + huge mux trees to LUTRAM + 256 valid FFs; the multi-minute optimization phase and the OOM disappear.

Verification: all 15 `tb_order_book_m2` and 4 `tb_tick_to_trade` cocotb tests pass under Verilator with cycle behaviour unchanged. (On a clean Verilator build pass `EXTRA_ARGS="-Wno-fatal"` to get past the pre-existing PINMISSING warning from Issue #4.)

Trade-off note: for true block RAM (`RAMB36`), convert all reads (the C/D/E lookup and the RESCAN stream) to synchronous and add read-wait pipeline stages; this adds one clock of read latency per access. Revisit if LUT pressure on the target device becomes a concern.

3. Issue #2 — Vivado native crash during synthesis [HIGH]

3.1 Symptom

An earlier (GUI) synthesis attempt aborted, leaving eight JVM/native crash dumps in the project root (`hs_err_pid75146.log`, `75251`, `75357`, `75462`, . . .).

3.2 Evidence

```
# An unexpected error has occurred (6) Aborted
...
libstdc++.so.6 ... _ZSt10unexpectedv
libxv_common.so ... hdi::tcltasks::detail::collection_wrapper_base::~...
libxv_synth.so ... HARTSInfo::periodicalDetect()
```

At GUI launch `vivado.log` reported `Available Virtual :  503 MB` — the machine was already memory-starved before the crash. Multiple helper synthesis processes aborted together (one dump per PID).

3.3 Root cause

A native abort (signal 6) inside the synthesis worker, occurring under severe memory pressure. Given the timing and the near-zero available virtual memory, this is most plausibly the GUI-side manifestation of Issue #1: the same order book FF/mux explosion drove the process out of memory and Vivado aborted rather than exiting cleanly.

3.4 Resolution / Status

Treat as resolved-by-#1: once the order book infers BRAM the memory pressure disappears. If native aborts persist afterward, capture a fresh `hs_err_pid` and investigate the Vivado install separately. The stale crash dumps can be deleted from the project root.

4. Issue #3 — XDC constraint errors in `top.xdc` [MEDIUM]

4.1 Symptom

Two CRITICAL WARNINGs every run while parsing the timing constraints, meaning some constraints are silently dropped.

4.2 Evidence

```
CRITICAL WARNING: [Designutils 20-1307] Command 'remove_from_collection' is
   not supported in the xdc constraint file. [rtl/top.xdc:31]
CRITICAL WARNING: [Common 17-1548] Command failed: can't read "in_ports":
   no such variable [rtl/top.xdc:32]
```

The offending lines:

```
# rtl/top.xdc : 30-32
set in_ports [remove_from_collection [all_inputs] [get_ports clk]]
set_input_delay -clock clk $IO_BUDGET $in_ports
```

4.3 Root cause

`remove_from_collection` is a Tcl/XDC command not permitted inside a pure `.xdc` constraint file. Line 31 therefore fails, `in_ports` is never set, and line 32 then fails with "no such variable" — so the input delay on all non-clock ports is never applied.

4.4 Resolution / Status (**fixed** 12 June 2026)

Replaced the `remove_from_collection` line in `rtl/top.xdc` with an XDC-legal `get_ports` filter that selects every input port except the clock:

```
set_input_delay -clock clk $IO_BUDGET \
    [get_ports * -filter {DIRECTION == IN && NAME != clk}]
```

This removes both CRITICAL WARNINGs and ensures the input-delay budget is actually applied (previously it was silently dropped).

Verification: the file was sourced under `tclsh` with the Vivado commands stubbed (and `remove_from_collection` stubbed to raise) — it now parses cleanly, whereas the old version errored on the unset `in_ports` variable. The two CRITICAL WARNINGs should be gone on the next synthesis run.

5. Issue #4 — Assorted synthesis warnings [LOW]

These do not block synthesis but are worth a review pass:

- **Set/reset same priority** — `[Synth 8-7137]` on several `itch_parser` registers (`msg_type_reg`, `timestamp_reg`, `order_ref_reg`, `side_reg`, `shares_reg`, `price_reg`; lines 139–144). Vivado warns this may cause sim/synth mismatch; clean up the reset coding style.
- **Unconnected ports** — `[Synth 8-7129]`: `m_axis_tready` on `tick_to_trade_top`, and the upper bits of `s_axil_wdata`/`s_axil_wstrb` on `latency_counter` have no load.
- **Constant-driven outputs** — `[Synth 8-3917]`: `m_axis_tdata[71:65]` tied to constant 0 (upper trade-output bits unused).

Action: confirm each is intentional (it is normal for unused AXI/AXIS sidebands during bring-up); otherwise wire up or trim the interface.

6. Environment notes

- Host: `abhishek-VMware-Virtual-Platform`, Ubuntu 24.04.2, AMD Ryzen 7 4800H, **18,412 MB** RAM, 12,884 MB swap.
- Tool: Vivado v2025.2 (64-bit), SW build 6299465.
- Target part: `xcu200-fsgd2104-2-e` (Alveo U200) — a large datacenter device; its part/timing database alone is memory-heavy to load, which compounds Issue #1.